



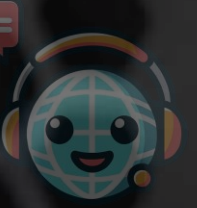
Developing LLM for Bangla Language

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TituLLMs: A Family of Bangla LLMs with Comprehensive Benchmarking (ACL 2025)



What is an LLM? Core Concepts

- Autoregressive Transformer trained to predict next token
- Capabilities emerge from scale + data diversity: reasoning, QA, summarization
- Compute–data–model scaling laws guide choices
- For low-resource languages, representation and tokenization dominate out



Why We had a plan to train a Base Model

- Available opensource/commercial models are not good at bangla language reasoning
- A strong base model is a high demand for finetuning chatbot and other downstreaming tasks
- Small models are also good for production uses considering latency



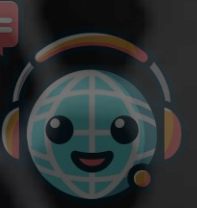
Why Build a Bangla-Centric LLM?

- Multilingual models underrepresent Bangla; byte-level tokenization harms efficiency
- Morphologically rich; colloquial + romanized usage is common
- High-quality Bangla web/book corpora are scarce; synthetic sources help
- Cultural nuance and exam-style knowledge require native datasets



TituLLMs in Brief

- Two sizes: 1B and 3B parameters (adapted from Llama-3.2)
- Pretraining corpus: ~37B Bangla tokens (268 GB cleaned)
- Extended tokenizer: ~48K new Bangla tokens; TPW drop from 7.84 → 1.90
- Context length: 4096; training ~1,750 H100 GPU hours



Pretraining Data Collection

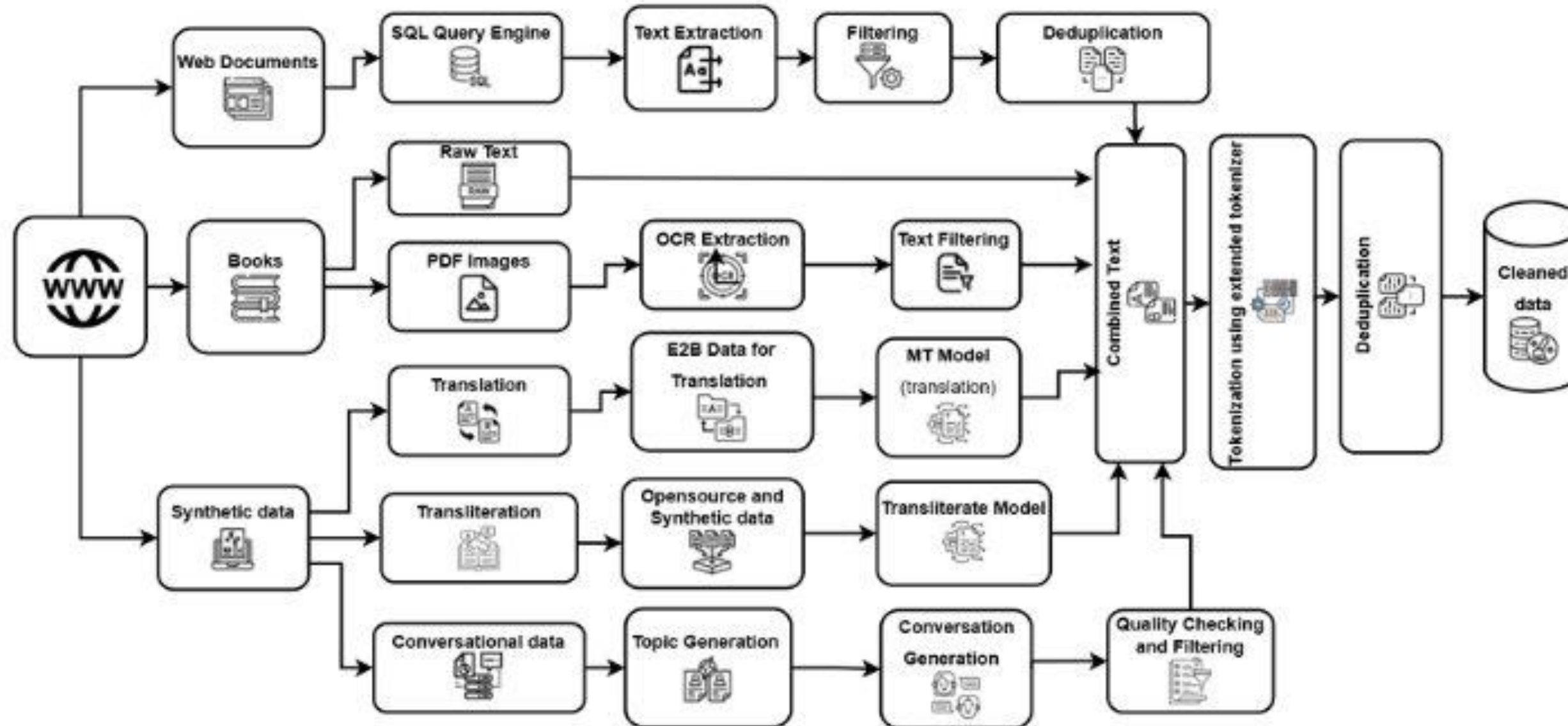


Pretraining Data Pipeline

- Sources: Web, Books, Synthetic, Sangraha
- Tools: AWS Athena (Common Crawl); Trafilatura for text extraction
- Quality: 18 handcrafted Bangla-specific filters + MinHash dedup
- Validation split preserves heterogeneity; ~22 GB held-out



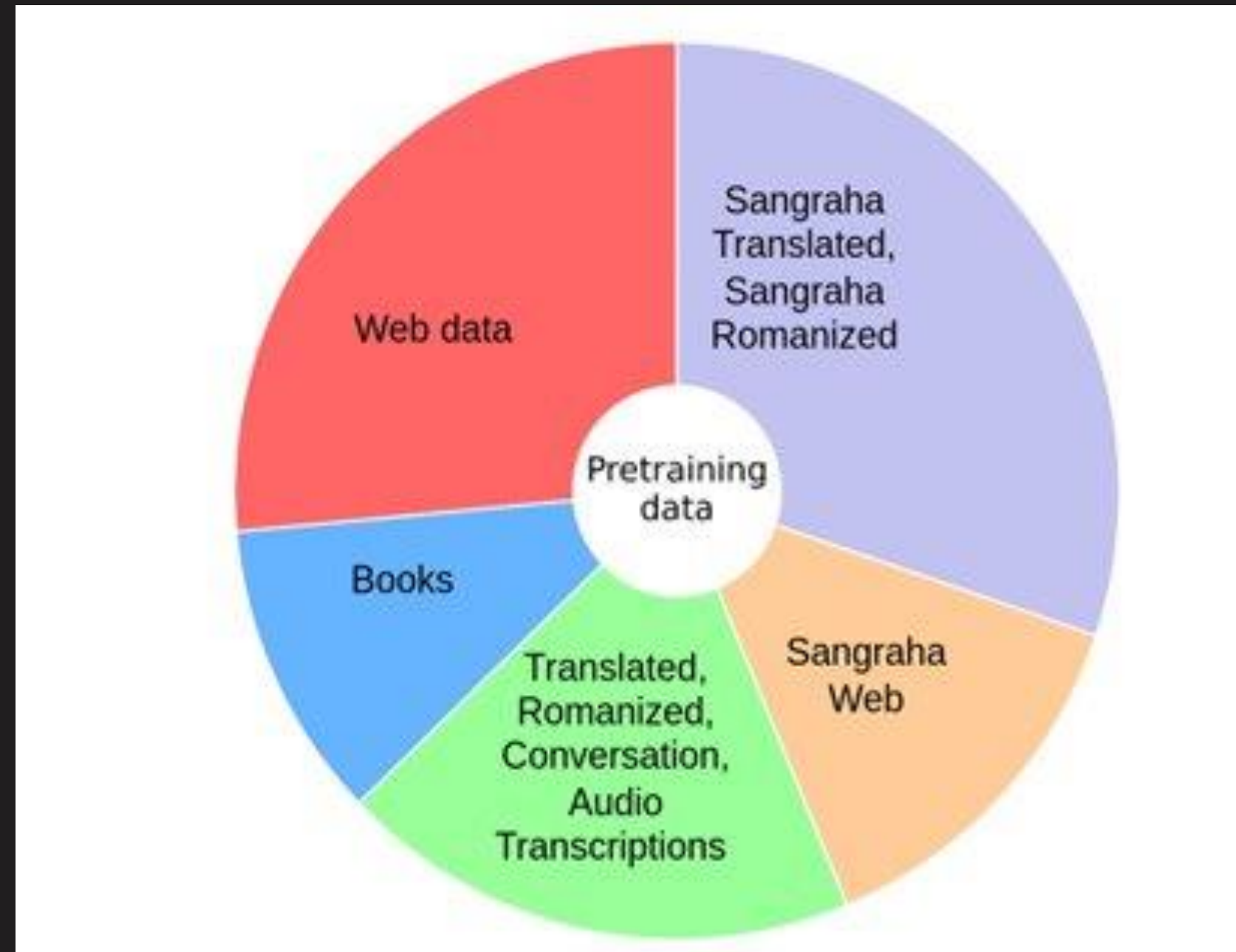
Data Pipeline





Dataset Composition (~37B Tokens)

- Web: ~9.8B; Books: ~4B
- Synthetic: ~7.06B (translated, romanized, conversation, ASR)
- Sangraha Web: ~5B; Sangraha Synthetic: ~10.94B
- Heterogeneous mix to cover formal + colloquial + romanized Bangla





Web Data: Collection & Cleaning

- Query Common Crawl via Athena; filter by language, .bd domains, host
- Extract clean text with Trafilatura; preserve structure; drop boilerplate
- Apply 18 Bangla-specific rules: sentence completeness, symbols, profanity, etc.
- Deduplicate with MinHash; thresholding to remove near-duplicates



Books: OCR & Post-processing

- Mix of digital PDFs and scanned images
- OCR: Google OCR + Tesseract; segmentation with YOLO for layout
- Filtering: KenLM page scores; word/sentence stats; % valid Bangla words
- Confidence gating: retain top ~50% of pages post-cleaning



Why Synthetic Data?

- Address scarcity of high-quality native Bangla content
- Cover colloquial speech, regional variants, romanized everyday usage
- Balance formal curricula/exam data with conversational style
- Boost reasoning and dialogue behaviors pre-instruction tuning



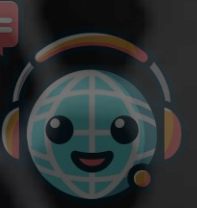
Synthetic Components & Scale

- ASR Transcripts: ~56k hours collected; ~1.3B tokens
- Agentic Conversations: ~1M dialogues; avg 8 turns
- Translation: ~60M en–bn pairs used for training a compact translator
- Romanized: large-scale transliteration via NLLB-based model
 - Used to create the transliteration dataset by selecting a small subset of collected Bangla Wikipedia articles.



Custom Translator for Pretraining

- NLLB-600M distilled backbone; trained on mixed open corpora
- In-house BLEU ≈ 37.6 ; strong SQM human scores across domains
- Used to translate English news/Wikipedia; added Sangraha translations
- Chosen for lightweight inference + Bangladeshi style alignment



Tokenizer Training



Why Base Tokenizer Fails for Bangla

- Llama tokenizer fragments Bangla into bytes/characters
- High Tokens per Word (TPW ≈ 7.84) $\rightarrow$ longer sequences, higher cost
- Loses morpheme/word boundaries; harms semantics and latency
- Need language-aware subword vocab to unlock efficiency



Training & Merging a New Tokenizer

- Sample 48 GB Bangla corpus; train multiple BPE tokenizers (32K–96K)
 - Using tiktoken library
- Merge with Llama-3.2 tokenizer to preserve base strengths
- Evaluate on 1 GB sample: pick config minimizing TPW vs vocab size
- Winner: plus-48K $\rightarrow$ TPW $\approx$ 1.90 (dramatic reduction)

Tokens per word (TPW) for different Llama models

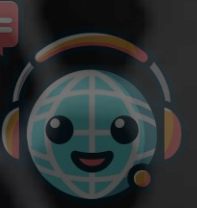


Model	TPW
Llama-3.2	7.8397
Llama-3.2-plus-32K	2.1346
Llama-3.2-plus-48K	1.9029
Llama-3.2-plus-64K	1.7946
Llama-3.2-plus-80K	1.7370
Llama-3.2-plus-96K	1.7034



Why Tokenizer Quality Matters

- Shorter sequences → lower memory & FLOPs for same context
- Better subword alignment → improved reasoning/local semantics
- Observed: boosts on CSQA/OBQA/PIQA without instruction tuning
- Also reduces inference latency for Bangla inputs



Benchmark Dataset



Why New Bangla Benchmarks?

- Existing evals skew to English/high-resource languages
- We need world knowledge, commonsense, and reading comprehension
- Create native + translated sets with quality controls
- Enable apples-to-apples comparisons for Bangla LLMs



Benchmark Suite ($\approx 132k$ items)

- Bangla MMLU ($\sim 87.9k$): exam-style multi-domain MCQs
- PIQA ($\sim 17.2k$): physical commonsense
- CommonsenseQA ($\sim 11k$): everyday commonsense
- OpenBookQA ($\sim 5.9k$): elementary science + reasoning
- BoolQ ($\sim 2k$): yes/no reading comprehension

Dataset	Method	Train	Val.	Test	Dev
Bangla MMLU	Manual	-	72,944	14750	175
BoolQ	GPT-4	815	432	729	-
CommonsenseQA	EST	9,741	1,221	-	-
OpenBookQA	EST	4,947	500	497	-
PIQA	EST	15,339	1,838	-	-



CSQA / OBQA / PIQA (EST-Translated)

- CSQA: commonsense with distractors
- OBQA: apply textbook facts to novel questions
- PIQA: practical physical reasoning choices
- All use EST to preserve idioms and avoid literal artifacts



BoolQ Bangla: Reading Comprehension

- Passages from Bangla Wikipedia/Banglapedia/news
- Questions/answers curated (yes/no); GPT-assisted generation
- Measures comprehension over realistic lengths
- Good probe for long-context weakness



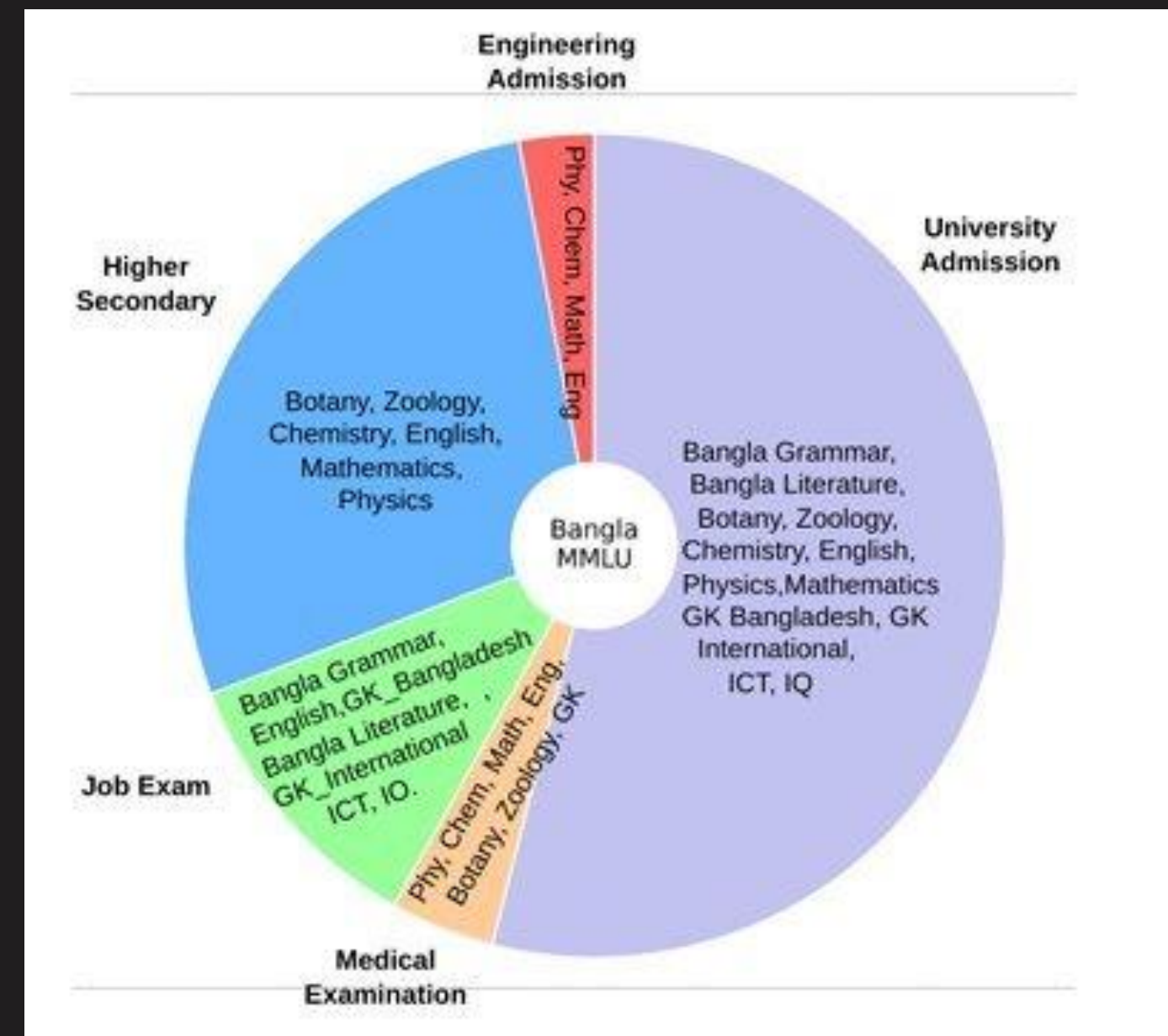
Bangla MMLU: Design & Coverage

- Sources: admission tests, job exams, medical, engineering
- Manual curation; category balance
- Tests world knowledge and exam heuristics
- Dominant share ensures robust signal for knowledge gaps



Bangla MMLU: Design & Coverage

University Admission (47,394)
Higher Secondary (25,437)
Job Exams (9,122)
Medical Admission (3,764)
Engineering Admission (2,152).





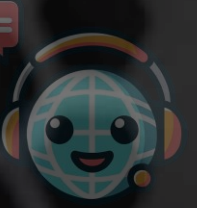
EST: Expressive Semantic Translation

- Pipeline: baseline MT → LLM refinements → candidate fusion → ranking
- Goal: idiomatic, context-faithful Bangla for benchmark creation
- Applied to CSQA, OBQA, PIQA; improves naturalness over literal MT
- Evaluated with BLEU/SacreBLEU/BERTScore + LLM grading



EST: Expressive Semantic Translation

Translator	BLEU	SBLEU	BS (F1)	LLM Score
Google	0.33	29.00	0.91	8.96
GPT-4o	0.26	22.05	0.90	8.91
IndicTrans2	0.27	23.74	0.90	8.73
Gemini-1.5-pro	0.31	27.08	0.89	8.80
EST	0.57	49.50	0.93	8.95



Pretraining



Model Selection

- Chinchilla Scaling Law : Optimal LLMs should use ~20 training tokens per parameter. <https://life architect.ai/chinchilla/>
- Trained both 1B and 3B models on ~37B tokens
- 1B model is slightly over the optimal point → better learning per parameter
- 3B model is somewhat undertrained vs optimal (~60B) but still learns efficiently with high-quality, Bangla-rich data



Model Adaptation (Llama-3.2 → TituLLMs)

- Extend embedding matrix for new vocab; reinit new token vectors
- Resize LM head; retain Transformer blocks/hyperparams
- Model sizes: 1B and 3B chosen for training/inference practicality
- Context length 4096; packing enabled



Pretraining Configuration

- Framework: LlamaFactory; 1 epoch over ~37B tokens
- Compute: ~1,750 H100 GPU hours
- Curriculum: mixed datasets; maintain heterogeneity
- Validation: normalized accuracy on held-out tasks; tokenizer TPW checks



Model Performance Overview (0/5-shot)

- Strengths: commonsense & applied reasoning (CSQA/OBQA/PIQA)
- Weaker: world knowledge (Bangla MMLU), long-context BoolQ
- Tokenizer upgrade correlates with reasoning gains
- Comparable or better than similar-size multilingual baselines



Results (normalized accuracy)

Model	S	BN MMLU	BoolQ	CSQA	OBQA	PIQA
davinci	0	0.30	0.53	0.22	0.30	0.52
	5	-	-	-	-	-
Llama-3.2-1b	0	0.28	0.53	0.23	0.32	0.53
	5	0.28	0.58	0.23	0.32	0.54
Llama-3.2-3b	0	0.33	0.53	0.26	0.32	0.57
	5	0.34	0.69	0.29	0.32	0.57
Gemma-2-2b	0	0.29	0.56	0.26	0.34	0.56
	5	0.32	0.60	0.28	0.33	0.56
Qwen-2.5-0.5b	0	0.30	0.53	0.21	0.31	0.54
	5	0.31	0.58	0.22	0.30	0.53
Qwen-2.5-1.5b	0	0.33	0.62	0.23	0.29	0.53
	5	0.35	0.68	0.23	0.30	0.52
SmolLM2-135m	0	0.23	0.53	0.22	0.31	0.52
	5	0.23	0.51	0.21	0.30	0.52
SmolLM2-360m	0	0.25	0.53	0.20	0.30	0.54
	5	0.24	0.52	0.21	0.29	0.53
SmolLM2-1.7b	0	0.29	0.53	0.22	0.31	0.53
	5	0.30	0.55	0.21	0.30	0.53
BLOOM-560m	0	0.23	0.53	0.26	0.31	0.54
	5	0.23	0.53	0.26	0.28	0.54
BLOOM-1b1	0	0.26	0.56	0.27	0.31	0.54
	5	0.23	0.58	0.27	0.31	0.55
BLOOM-1b7	0	0.27	0.53	0.27	0.32	0.55
	5	0.27	0.59	0.30	0.31	0.56
BLOOM-3b	0	0.26	0.53	0.27	0.33	0.58
	5	0.23	0.53	0.32	0.31	0.58
BongLLaMA-3.2-1b	0	0.25	0.53	0.22	0.33	0.52
	5	0.26	0.53	0.24	0.31	0.53
BongLLaMA-3.2-3b	0	0.30	0.53	0.21	0.27	0.51
	5	0.33	0.54	0.20	0.29	0.50
TituLLM-1b-v2.0	0	0.25	0.53	0.26	0.32	0.58
	5	0.25	0.51	0.28	0.33	0.57
TituLLM-3b-v2.0	0	0.25	0.53	0.28	0.32	0.58
	5	0.25	0.54	0.33	0.35	0.60



Reasoning Results Highlights

- TituLLM-3B 5-shot: CSQA ≈ 0.33 ; OBQA ≈ 0.35 ; PIQA ≈ 0.60
- Gains vs 0-shot on CSQA/OBQA suggest few-shot adaptability
- Outperforms many $\leq 3B$ baselines on these tasks



Why the Gains? (Hypotheses)

- Tokenizer delivers word-level units → better semantics
- Synthetic conversations + romanized data enrich everyday reasoning
- Science & physical commonsense represented in pretraining
- Even 1 epoch sufficed to shift behavior in target language



Known Limitations

- Single-epoch pretraining on ~37B tokens (under-trained vs SOTA)
- Long-context understanding remains weak (BoolQ plateau)
- Bangla-only pretraining limits global/world knowledge
- No instruction tuning yet → limited task-following



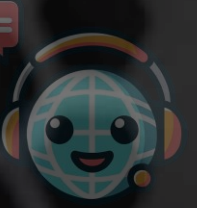
Key Takeaways

- Data quality + tokenizer design are make-or-break for low-resource LLMs
- Merging a Bangla-aware tokenizer slashes TPW and boosts reasoning
- Synthetic pipelines (ASR, EST, romanized, agentic convos) fill crucial gaps
- Benchmarks must reflect native knowledge and culture



Dataset and Model

- **Dataset**
 - **Pretraining:** [hishab/titulum-bangla-corpus](#) · Datasets at Hugging Face
 - **Benchmarking:**
 - **MMLU Dataset:** [hishab/titulum-bangla-mmlu](#) · Datasets at Hugging Face
 - **commonsenseqa-bn:** [hishab/commonsenseqa-bn](#) · Datasets at Hugging Face
 - **openbookqa-bn:** [hishab/openbookqa-bn](#) · Datasets at Hugging Face
 - **hishab/piqa-bn:** [hishab/piqa-bn](#) · Datasets at Hugging Face
 - **hishab/boolq_bn:** [hishab/boolq_bn](#) · Datasets at Hugging Face
- **Model**
 - **3B:** [hishab/titulum-llama-3.2-3b-v2.0](#) · Hugging Face
 - **1B :** [hishab/titulum-llama-3.2-1b-v2.0](#) · Hugging Face



Question and Answer